

THieF Covid-19 Example

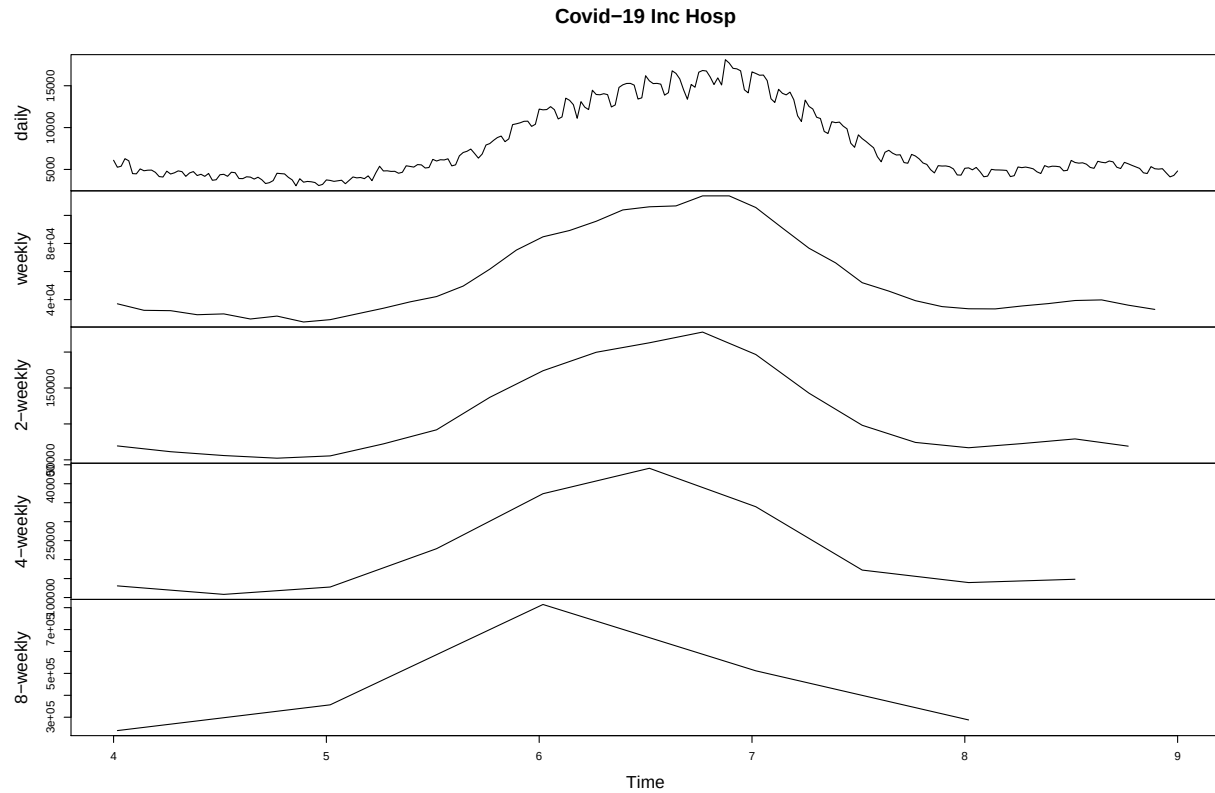
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```
## Error in assign(generic, get(generic, asNamespace("greybox")), envir = asNamespace(pkg)) :  
##   cannot change value of locked binding for 'forecast'
```

Example with Covid-19 inc hosp data. We adjust the top aggregation level to be 8 weeks, as we can't reasonably predict anything with a longer horizon well-enough.

```
hosp_truth <- load_truth("HealthData",  
  "inc hosp",  
  temporal_resolution="weekly",  
  data_location = "remote_hub_repo") %>%  
  dplyr::filter(target_end_date >= as.Date("2020-06-28"),  
    target_end_date <= as.Date("2021-06-27"),  
    location == "US") %>%  
  mutate(day = wday(target_end_date), epi_week = epiweek(target_end_date)) %>%  
  dplyr::select(value, epi_week, day)  
  
ht_day <- hosp_truth$value  
ht_day_ts <- ts(ht_day, start = c(2020, 179), end = c(2021, 178), frequency = 365)  
  
# create list of aggregation levels  
agg_list <- list(56, 42, 28, 14, 7, 1)  
  
# Construct aggregates (from weekly data)  
ht_day_ts_ <- ts(ht_day, start = c(4,1), end = c(9, 1), frequency = 56)  
# where epiweek 1 of 2020 is 1  
day_agg_ <- tsaggregates(ht_day_ts_, m = 56, aggregatelist = agg_list)  
agg.names <- c("daily", "weekly", "2-weekly", "4-weekly", "8-weekly")  
for(i in seq_along(day_agg_))  
  names(day_agg_)[[i]] <- agg.names[i]  
plot(day_agg_, main="Covid-19 Inc Hosp")
```



```
# Compute base forecasts
base_fc_day <- list()
for(i in seq_along(day_agg_))
  base_fc_day[[i]] <- forecast(auto.arima(day_agg_[[i]]), h=2*frequency(day_agg_[[i]]))

# Reconcile forecasts
reconciled_fc_day <- reconcilethief(base_fc_day, aggregatelist = agg_list)

# Plot forecasts before and after reconcilliation
par(mfrow=c(2,3))
for(i in seq_along(base_fc_day))
{
  plot(reconciled_fc_day[[i]], main=agg.names[i],
       ylim = c(0, max(reconciled_fc_day[[i]]$x, reconciled_fc_day[[i]]$upper)))
  lines(base_fc_day[[i]]$mean, col='red') # plots red mean line
}
```

